

NHS Expenditure Analysis - Cheat Sheet

1. Key Statistical Tests & Their Interpretations

1.1 Welch's t-Test (Unequal Variance t-Test)

- **Purpose:** Compares the means of two groups (pre-pandemic vs. post-pandemic spending) when variances are unequal.
- **Interpretation:**
 - $p < 0.05$ → Statistically significant difference in spending before and after COVID-19.
 - $p > 0.05$ → No strong evidence of a shift in spending.
- **Example:**
 - **Infectious Diseases Total:** $p = 0.0624$ (Borderline significant increase).
 - **Musculo Skeletal System Problems Total:** $p = 0.011$ (Statistically significant decrease).

1.2 Levene's Test (Variance Test)

- **Purpose:** Determines if spending variance changed significantly over time.
- **Interpretation:**
 - $p < 0.05$ → Significant difference in spending volatility (more unpredictable funding patterns).
 - $p > 0.05$ → Variability remained stable.
- **Example:**
 - **Infectious Diseases Total:** $p = 0.0124$ (Increased volatility post-pandemic).
 - **Musculo Skeletal System Problems Total:** $p = 0.982$ (Stable spending volatility).

1.3 Correlation Analysis

- **Purpose:** Examines relationships between different NHS expenditure categories.
- **Interpretation:**
 - **Positive correlation (>0.5)** → Spending in one category is linked to another.
 - **Negative correlation (<-0.5)** → When spending in one category rises, another declines.

2. Visualization Techniques & When to Use Them

Visualization	Purpose	Key Insights
Line Chart	Trend analysis	Shows changes in spending over time (total & per capita).
Stacked Area Chart	Category-wise expenditure evolution	Reveals how NHS priorities shifted over time.
Heatmap	Budget shifts over time	Highlights major funding increases/decreases.
Grouped Bar Chart	Pre- vs. Post-Pandemic Spending	Compares budget shifts between two time periods.

Visualization	Purpose	Key Insights
Boxplot	Spending volatility analysis	Shows variance in budget allocations by category.
Correlation Heatmap	Identifies category relationships	Determines how NHS spending in one category affects others.

3. Key Findings & Their Significance

3.1 Total NHS Expenditure Trends

- NHS spending consistently increased from **2009-2023**.
- **Biggest growth in 2020-21** (COVID-19 emergency response).
- Spending remained **higher than pre-pandemic levels**, suggesting **lasting structural shifts**.

3.2 Budget Reallocation Trends

- **Biggest Increase:** Infectious Diseases Total (pandemic-driven response).
- **Biggest Decrease:** Musculo Skeletal System Problems (possible deprioritization).

3.3 Efficiency Analysis

- **Total vs. Per Capita Growth Aligned** → Indicates NHS efficiently scaled with population needs.
- **No major increase in spending volatility** → Suggests controlled financial planning.

4. Anticipated Interview Questions & Suggested Responses

4.1 Why Did Infectious Disease Funding Increase?

Answer: The **COVID-19 pandemic required emergency funding** for vaccines, ICU expansions, and testing. While funding has since declined, it remains above pre-pandemic levels, suggesting a long-term shift in NHS priorities.

4.2 Why Did Musculo Skeletal System Funding Decrease?

Answer: Possible factors include **budget reallocation to more urgent areas (e.g., mental health, emergency care)** or efficiency improvements in treatment models. The statistical significance confirms a **deliberate and lasting budget reduction**.

4.3 How Did You Ensure Statistical Rigor?

Answer:

- Used **Welch's t-Test** for spending shifts (accounts for different sample sizes pre/post-pandemic).
- Used **Levene's Test** for variance (ensured budget fluctuations were measured accurately).
- Ensured visualizations captured key insights beyond numerical analysis.

4.4 What Insights Can Policymakers Take From This?

Answer:

- NHS should **monitor long-term effects** of post-pandemic budget reallocations.
 - **Infectious disease preparedness should remain a priority** to prevent future health crises.
 - Musculo Skeletal System funding cuts should be examined **to assess potential care gaps**.
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This cheat sheet serves as a **quick reference guide** to help answer **interview questions with confidence** while reinforcing the statistical significance of the analysis. 🚀